NEST: Locally Certified Search for Structural-Entropy Hierarchies

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Abstract. Structural entropy evaluates a graph hierarchy through the description length of an encoding tree. HCSE and BBM [12] construct such trees but do not determine whether the result admits a local entropy-reducing topology change. We introduce NNI-certified Entropy Search over Trees (Nest), a multi-start optimizer based on rooted nearest-neighbor interchange (NNI). For weighted graphs, we derive the exact NNI entropy change and prove that an improving rotation can be inaccessible to a monotone merge or compression. The formula supports best-improvement descent and a checkable one-NNI local-optimality certificate. Because a better tree may require a temporary increase, Nest also tests bounded two-move paths but commits only improving endpoints. We also formulate an $O(3^n)$ subset dynamic program, using binary sufficiency [14, 12], to compute exact global optima for audit graphs. Across 500 graphs from five 64-vertex hierarchical regimes, Nest lowers entropy by 0.4239±0.0086 bits relative to the better of HCSE and BBM and wins all 500 paired comparisons. In sealed, candidate-budget-matched audits, 32 coalescent starts reach the exact global optimum on 249/250, 99/100, and 22/25 instances at $n = 12, 14, 16$, respectively (370/375 overall), whereas 32-call HCSE and label-free BBM reach 2/375 and 1/375. The exact audits show that the local certificate and bounded escape together recover global optima reliably across the tested regimes and sizes.

Keywords: Structural entropy · hierarchical clustering · tree editing · nearest-neighbor interchange · graph algorithms

1 Introduction

Many networks are organized at more than one scale: vertices form local groups, related groups form larger modules, and the desired output is therefore a tree rather than one flat partition. Structural entropy supplies an information-theoretic objective for such an encoding tree [7]. Recent HCSE and BBM methods construct low-entropy hierarchies through recursive partitioning and balanced merging [12]. Their construction rules leave a complementary question unanswered: *after construction, can a local topology change further reduce structural entropy?*

A hierarchy constructor can make an early merge that later steps never revisit, even when rotating a nearby internal edge would lower structural entropy. We therefore seek a topology operator whose effect can be computed exactly, whose termination certifies the inspected neighborhood, and whose output can be checked against a true optimum on exactly solvable instances.

Nearest-neighbor interchange (NNI) is the elementary rotation of a rooted binary tree. It changes one internal edge while preserving every leaf. NNI is standard in evolutionary-tree search and has been studied for other hierarchical-clustering objectives, but its structural-entropy change has not been used as an exact graph-hierarchy optimizer. We show that, for $((A, B), C) \to (A, (B, C))$, all objective terms cancel except two cross-module edge weights and three module volumes. This identity yields monotone descent and a one-NNI local-optimality certificate.

Our method, NNI-certified Entropy Search over Trees (Nest), refines several inexpensive initial trees with the exact operator and returns the lowest independently verified endpoint. A bounded two-move beam can cross a small intermediate barrier, but accepts a pair only when its endpoint lowers the full objective. A randomized-restart variant, Nest-R, increases basin coverage without inspecting labels or the exact optimum. Using prior binary sufficiency [14, 12], we specialize an exact dynamic program over vertex subsets to compute global optima for small-graph audits. The resulting pipeline connects an exact local identity, a certified search procedure, and an independent global benchmark.

Our contributions are:

- an exact weighted-graph NNI delta, a proof of separation from monotone single merge/compress steps, and a checkable one-NNI local certificate;
- Nest, which combines multi-start construction, exact NNI refinement, and bounded two-move escape without weakening the output guarantee;
- an SE-specific $O(3^n)$ arbitrary-subset trellis optimizer for small graphs, built on prior binary sufficiency, and an exact audit of Nest's optimality gap; and
- a paired evaluation against SE agglomeration, Louvain-2L, Paris, and the official HCSE and BBM implementations, with entropy, hierarchy recovery, runtime, memory, operator audits, and component ablations.

Across five frozen regimes, Nest lowers entropy by 0.5468 ± 0.0218 bits versus HCSE and 0.5213 ± 0.0101 bits versus BBM, winning every paired comparison across 500 graphs. Across sealed exact suites at $n = 12, 14, 16$, 32 coalescent Nest starts reach the global optimum in 370/375 cases, compared with 2/375 for 32-call HCSE and 1/375 for 32-call label-free BBM.

2 Preliminaries

Weighted graph and encoding tree. Let $G = (V, E, w)$ be an undirected graph with nonnegative edge weights. For $S \subseteq V$, let $V_S = \mathrm{vol}(S) = \sum_{u \in S} d_u$, and let g_S be the total weight of edges with exactly one endpoint in S. The total graph volume is $\mathrm{vol}(V) = \sum_u d_u = 2 \sum_{e \in E} w_e$. An encoding tree T has one leaf per vertex. Each internal node a represents the union S_a of its descendant leaves; a^- denotes its parent.

The structural entropy of G under T is

$$H^T(G) = -\sum_{a \neq r} \frac{g_{S_a}}{\mathrm{vol}(V)} \log_2 \frac{V_{S_a}}{V_{S_{a^-}}}. \tag{1}$$

Our sole optimization objective is Eq. (1); lower is better. No ground-truth label is used to construct or select a tree. For a zero-volume module, its summand is defined as zero; such a module has no incident positive-weight edge and can be attached after optimizing the positive-volume part of the tree.

Constructors. We distinguish a *constructor*, which produces an initial tree, from a *refiner*, which searches neighboring topologies. Our candidate pool contains (i) a binary SE-agglomerative dendrogram, (ii) a recursive SE partition tree, and (iii) a Paris dendrogram when the optional implementation is available [2]. HCSE and BBM are distinct external constructors and are evaluated from their official implementation [12].

Rooted NNI. An NNI move acts on a binary parent–child edge. Here A, B, and C denote three disjoint subtrees, and $P = A \cup B \cup C$ denotes their shared local parent module: the set of all leaves contained in those three subtrees. Writing the local topology as $((A, B), C)$, the two nontrivial alternatives are $(A, (B, C))$ and $(B, (A, C))$. The leaf set, graph, and all modules outside P remain unchanged. A rooted binary tree with n labeled leaves has finitely many topologies, namely $(2n - 3)!!$, so any strictly decreasing local search terminates.

3 NNI-Certified Entropy Search

Nest builds several initial hierarchies, improves each through NNI rotations, and returns the tree with the lowest structural entropy. Equation (2) scores every candidate rotation exactly, while a full recomputation of Eq. (1) checks every accepted tree. An exact dynamic program supplies the global optimum for the small-graph audits.

3.1 A local structural-entropy identity

For disjoint sets X, Y, write $W(X, Y) = \sum_{u \in X, v \in Y} w_{uv}$. Consider the rotation shown in Fig. 1. The disjoint subtrees A, B, C fill their local parent $P = A \cup B \cup C$; the rotation replaces its child module $X = A \cup B$ with $Y = B \cup C$ while leaving the rest of the hierarchy unchanged.

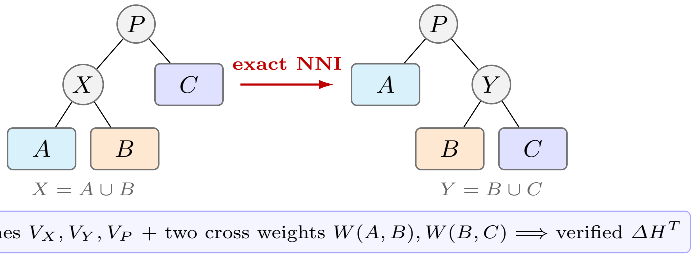

Fig. 1. A rooted NNI replaces only $X = A \cup B$ by $Y = B \cup C$. The highlighted local statistics determine the exact full-objective change; all leaves and the surrounding tree remain fixed.

Theorem 1 (Exact weighted NNI delta). *For an eligible rotation with $V_X, V_Y, V_P > 0$, the change in tree structural entropy is*

$$\Delta H^T = \frac{2}{\mathrm{vol}(V)} \left[W(A,B) \log_2 \frac{V_P}{V_X} + W(B,C) \log_2 \frac{V_Y}{V_P} \right]. \tag{2}$$

The symmetric alternative follows by exchanging A and B.

Proof. Deferred to Appendix A.

Proposition 1 (Atomic advantage over merge–compress). *Let T_c contract the edge from P to $X = A \cup B$, exposing A, B, C as siblings, and let T' then merge B, C under $Y = B \cup C$. For these HCSE compress and merge primitives,*

$$H^{T_c}(G) - H^T(G) = \frac{2W(A,B)}{\mathrm{vol}(V)} \log_2 \frac{V_P}{V_X} \geq 0, \tag{3}$$

$$H^{T'}(G) - H^{T_c}(G) = -\frac{2W(B,C)}{\mathrm{vol}(V)} \log_2 \frac{V_P}{V_Y} \leq 0. \tag{4}$$

Their sum is Eq. (2). Thus NNI lowers entropy when the merge gain in Eq. (4) exceeds the compression loss in Eq. (3), although neither desired reparenting nor an improving compression is available as one primitive.

Proof. Deferred to Appendix A.

This proves strict separation from monotone single-primitive search, not global dominance over a complete HCSE round, which may cross an intermediate increase. We evaluate Eq. (2) from cached sets and sparse adjacency lists, then verify every selected move by fully recomputing Eq. (1).

3.2 One-step certificate and compound escape

Algorithm 1: Exact NNI refinement of one candidate tree

Input: Graph G, tree T, tolerance ε, barrier β, beam b

repeat
  evaluate Eq. (2) for every legal NNI of T;
  apply the most negative move and verify the full objective;
until *no improving one-step move exists*;
for *at most R compound rounds* **do**
  retain the b first moves with smallest ΔH and $\Delta H \leq \beta$;
  expand every legal second NNI from each retained intermediate tree;
  if *no two-move endpoint has entropy below $H^T(G)$* **then**
    break;
  commit the best endpoint; rerun one-step descent;
return T

Proposition 2 (Monotonicity and local certificate). *One-step refinement never increases $H^T(G)$. If it stops before its move budget, the returned tree has no improving rooted NNI on any eligible binary parent–child edge.*

Proof. Deferred to Appendix A.

Strict one-step descent can stop before a better tree separated by a shallow barrier. Algorithm 1 therefore keeps a short list—a beam—of the b most promising first moves whose increase is at most β, then evaluates a second NNI from each intermediate tree. It never commits the intermediate tree alone.

Proposition 3 (Safe compound search). *The compound stage never worsens its input. It can reach an improving endpoint even when every one-step neighbor is non-improving.*

Proof. Deferred to Appendix A.

3.3 Fast NNI-certified entropy search

Nest applies Algorithm 1 independently to SE-agglomerative, recursive-SE, and Paris starts and returns the candidate with minimum verified entropy. The initializer is recorded for every output, so gains can be attributed to candidate selection, one-step refinement, and compound escape. With n leaves there are at most $2(n-2)$ oriented NNI alternatives per sweep. Our implementation uses best improvement, cached leaf sets, sparse cross-weight scans, and an $O(m \log h + n)$ full verification only for a selected move, where h is tree height. We make runtime claims empirically rather than assuming balanced trees.

Randomized basin coverage. Nest-R adds q coalescent starts, each built by repeatedly merging two uniformly selected current components, then applies the same NNI routine. The distribution is not uniform over labeled topologies. Selection uses only rescored entropy; increasing q cannot worsen the result, and its global-hit rate is measured against exact optima below.

3.4 Exact global audit on small graphs

To measure how often certified endpoints coincide with global optima, we use a separate exact solver. SuperTAD proves binary sufficiency for general graphs, and HCSE also notes it [14, 12]; we restate the result to make the audit self-contained.

Proposition 4 (Binary sufficiency). *Every encoding tree has a binary refinement whose structural entropy is no greater. Consequently, a global minimum exists among rooted binary trees.*

Proof. Included for completeness in Appendix A.

For a nonempty vertex subset S, let $F(S)$ be the minimum contribution of a binary subtree rooted at S, excluding the edge from S to its parent. Then

$$F(\{v\}) = 0,$$
$$F(S) = \min_{A \dot{\cup} B = S} \left\{ F(A) + F(B) - \frac{g_A}{\mathrm{vol}(V)} \log_2 \frac{V_A}{V_S} - \frac{g_B}{\mathrm{vol}(V)} \log_2 \frac{V_B}{V_S} \right\}. \tag{5}$$

Enumerating each unordered split once gives $O(3^n)$ time and $O(2^n)$ memory, an SE-specific subset dynamic program, equivalently a trellis whose states are vertex subsets [9]. By Proposition 4, $F(V)$ is the global minimum over all encoding trees, not only over the binary subclass. The exact audits apply this routine at $n = 12, 14, 16$; the deployable Nest algorithm does not invoke it.

For an additional graph-only certificate, an edge-LCA expansion gives

$$H^T(G) = \sum_{uv \in E} \frac{w_{uv}}{\mathrm{vol}(V)} \log_2 \frac{V_{\mathrm{lca}(u,v)}^2}{d_u d_v} \geq \sum_{uv \in E} \frac{w_{uv}}{\mathrm{vol}(V)} \log_2 \frac{(d_u + d_v)^2}{d_u d_v}. \tag{6}$$

The inequality follows from $V_{\mathrm{lca}(u,v)} \geq d_u + d_v$. It is a universal lower bound, but the exact dynamic program gives the sharper certificates reported below.

4 Experiments

4.1 Questions and protocol

The experiments test the consequences of the exact local identity and the search procedure derived from it. We ask: (Q1) how often do established constructors leave an improving NNI; (Q2) does compound search add improvements beyond a one-step certificate; (Q3) does lower entropy preserve or improve recovery of a planted hierarchy; and (Q4) how close is Nest to the global optimum on exactly solvable graphs?

The sealed main protocol uses 100 graph seeds per regime. Every method sees the identical weighted graph, and every tree is rescored by the same independent implementation of Eq. (1). We report paired means and 95% t-intervals, raw per-seed records, wall-clock time, and Python allocation peak. The compound search uses beam width 16, barrier $\beta = 0.05$ bits, and at most eight accepted compound rounds. No label is used by Nest for model selection. BBM is additionally shown with the planted number of fine clusters, which is an oracle advantage and is marked as such.

Data. Our controlled suite contains five two-level hierarchical stochastic block models: clean, noisy, imbalanced, weighted, and weak-hierarchy. Each balanced graph has 64 vertices, eight fine blocks, and four coarse blocks; the imbalanced regime has the same total size with block sizes from 4 to 12. The generator manifest, any 0.01-weight connectivity edges, and every seed are stored with the results. This family follows the hierarchical stochastic block model used to study recovery beyond worst-case inputs [3].

Methods. We compare SE agglomeration, a two-level Louvain tree [1], Paris [2], official HCSE, official BBM with oracle k [12], and NEST. We also refine each external constructor independently to measure its one-step and compound gap rather than crediting all gains to our preferred initializer.

Metrics. The primary metric is $H^T(G)$. Dendrogram purity is computed for the planted fine and coarse partitions: for every same-class vertex pair, it measures the class fraction under the pair's predicted LCA, then averages over pairs. We also report the fraction of runs improved by one-step and compound NNI, number of moves, runtime, and memory. For the exact suite we report the additive gap $H^T_{\mathrm{NEST}} - H^*$, relative gap $(H^T_{\mathrm{NEST}}/H^* - 1)100\%$, and *optimal-hit rate*: the fraction of graph instances whose objective is within $\max(10^{-9}, 10^{-8}|H^*|)$ of the exact optimum. This objective metric measures optimization accuracy independently of planted-tree recovery.

4.2 A strict one-step trap

Before aggregate evaluation, an eight-vertex weighted witness verifies the purpose of compound search. Exact one-step descent stops at 1.970038 bits. Every one-NNI delta is nonnegative, but a first move crossing a 0.013970-bit barrier followed by a second move and ordinary descent reaches 1.920593 bits, a 0.049445-bit reduction. This witness is generated from a fixed seed and is covered by a regression test.

4.3 Frozen benchmark results

Table 1 reports the raw output of each method, before applying our audit to competing trees. NEST has the lowest mean H^T in all five regimes and on 500/500 individual paired graphs. Its paired reduction is 0.5468 ± 0.0218 bits relative to HCSE and 0.5213 ± 0.0101 bits relative to BBM; it is lower on all 500 graphs against each method. Figure 2 shows paired evidence against the stronger of HCSE and BBM on each graph.

4.4 The operator audit

The constructor audit asks whether the exact local operator can improve a returned tree. Table 2 and Fig. 3 summarize the improvement frequency, entropy reduction, and runtime tradeoff.

Table 1. Tree structural entropy H^T over 100 graphs per regime (mean ± 95% CI; lower is better). BBM† receives the planted fine-cluster count.

Method	Clean	Noisy	Imbal.	Weighted	Weak
SE–agglomerative	2.899 ± 0.021	3.723 ± 0.017	3.162 ± 0.023	2.445 ± 0.016	3.461 ± 0.022
Louvain–2L	4.109 ± 0.022	4.603 ± 0.013	4.241 ± 0.019	3.657 ± 0.032	4.432 ± 0.018
Paris	2.869 ± 0.022	3.709 ± 0.017	3.129 ± 0.023	2.349 ± 0.015	3.438 ± 0.023
HCSE	3.228 ± 0.031	4.485 ± 0.068	3.598 ± 0.036	2.741 ± 0.029	4.032 ± 0.051
BBM†	3.335 ± 0.021	4.118 ± 0.020	3.573 ± 0.025	3.007 ± 0.020	3.924 ± 0.022
NEST (ours)	**2.827 ± 0.021**	**3.683 ± 0.017**	**3.086 ± 0.023**	**2.340 ± 0.015**	**3.414 ± 0.021**

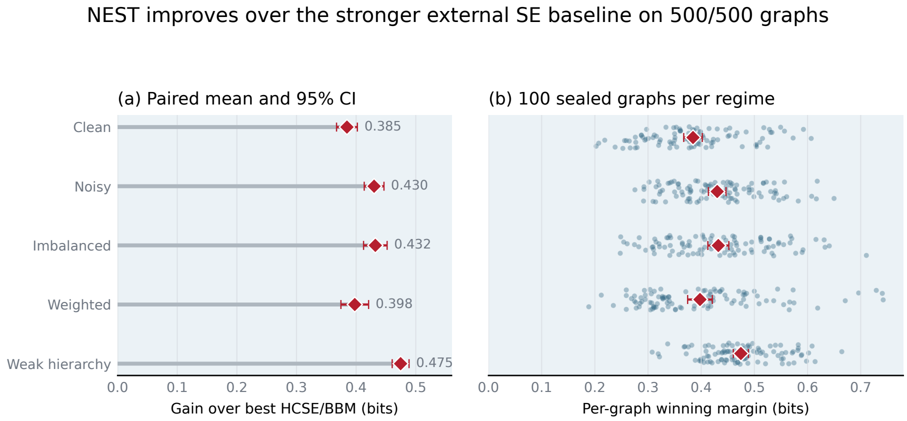

Fig. 2. Paired advantage over external structural-entropy methods. For each graph, the reference is the lower-entropy result of HCSE and BBM. (a) Regime-level mean reduction and 95% paired confidence interval. (b) Per-seed margins with small vertical jitter for visibility. NEST has lower entropy on all 500 graphs.

One-step NNI improves every SE-agglomerative and oracle-BBM tree. After the one-step certificate, compound search further improves 62% and 93% of those trees, respectively. For Paris, the one-step and compound rates are 74% and 41%; for HCSE they are 3% and 4%. The result shows that constructor quality and NNI-local optimality measure different properties of a hierarchy.

4.5 Components and exact optimality

NEST forms a small candidate set from SE agglomeration, recursive SE, and Paris, refines each with the same exact operator, and returns the lowest independently rescored endpoint. The main benchmark contains 3,500 method records and 500 generator manifests. Its verifier checks hashes, schema, paired seeds, unique keys, and monotonicity of committed NNI endpoints.

On a separate frozen 50-graph subset, candidate pooling improves 34 graphs, exact one-step NNI improves 45, and compound search adds gains on 30. Table 3 shows that each component contributes to the final result.

Table 2. NNI audit pooled over 500 paired graphs. Reduction is relative to each constructor's raw H^T; rates are fractions of runs improved.

Constructor	1-NNI gain	1-NNI hit	2-step gain	2-step hit	Time
SE–agglomerative	1.97%	100%	0.14%	62%	0.642s
Paris	0.07%	74%	0.06%	41%	0.388s
HCSE	0.00%	3%	0.00%	4%	0.108s
BBM†	14.02%	100%	0.46%	93%	1.962s

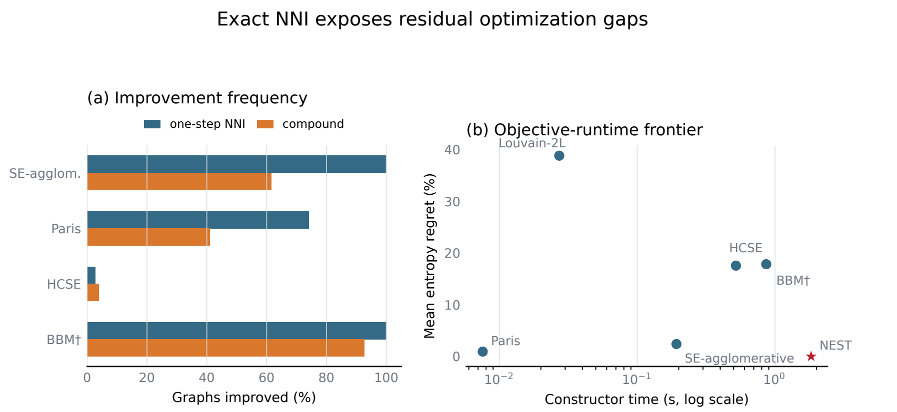

Fig. 3. Operator diagnostics pooled over 500 paired graphs. (a) Fraction of external constructor outputs improved by one-step NNI and by compound search after the one-step certificate. (b) Mean entropy regret versus constructor runtime; lower-left is better. Runtime includes Python allocation tracing and should be interpreted comparatively.

For the global audit, disjoint calibration fixes $q = 32$, after which we seal 250 graphs at $n = 12$, 100 at $n = 14$, and 25 at $n = 16$ across the same five regimes. Every restarted method receives 32 candidates. Selection uses only structural entropy; the exact dynamic program is consulted afterward. Table 4 shows that NEST reaches 370/375 optima, versus 2/375 for HCSE and 1/375 for label-free BBM. The table reports exact binomial intervals at every graph size. All completed blocks have status zero, matching SHA-256 seals, complete seed coverage, finite outcomes, and independently recomputed summaries. The combined result is 370 exact optima across 375 independently generated audit graphs.

4.6 Real networks

Table 5 complements the controlled suites with four networks bundled by NetworkX or the artifact library. NEST obtains the lowest raw entropy on 4/4networks. This experiment measures objective minimization on all four networks and uses label-based evaluation where a standard label is available.

Table 3. Component ablation pooled over 50 paired graphs. The gain and improved-run columns compare each row to the preceding row.

Variant	Mean H^T	Mean gain	Improved runs
SE agglomeration	3.1435	–	–
+ candidate pool	3.1041	0.0394	34/50
+ exact 1-NNI	3.0794	0.0247	45/50
+ compound escape	3.0759	0.0035	30/50

Table 4. Sealed exact-optimum audit with 32 candidates per method. Parentheses give exact 95% Clopper–Pearson intervals for the Nest hit rate. Gap is relative to H^*.

n	Graphs	Nest exact	95% CI	HCSE exact	BBM exact	Mean / max gap
12	250	**249/250**	[97.79, 99.99]%	2/250	1/250	0.00032% / 0.080%
14	100	**99/100**	[94.55, 99.97]%	0/100	0/100	0.00121% / 0.121%
16	25	**22/25**	[68.78, 97.45]%	0/25	0/25	0.03990% / 0.449%

Table 5. Tree entropy on four bundled real networks (lower is better). BBM uses the ground-truth k when labels exist and Louvain's k otherwise.

Method	Karate	Florent.	Les Mis.	Davis
SE–agglomerative	2.698	1.923	3.078	3.055
Louvain–2L	3.405	2.452	3.923	3.813
Paris	2.597	1.923	2.920	2.981
HCSE	2.929	2.172	3.930	3.206
BBM	3.330	2.254	3.199	3.387
Nest (ours)	**2.586**	**1.919**	**2.916**	**2.962**

5 Related Work

Structural entropy and hierarchy construction. Li and Pan introduced structural information as an encoding-tree measure of network organization [7]. HCSE and BBM develop hierarchy construction and balancing operations for this objective [12]. SuperTAD proves binary sufficiency and uses it in contiguity-constrained dynamic programs [14]; HCSE also notes the property [12]. Our exact audit instead optimizes arbitrary vertex subsets. The structural-entropy survey organizes the broader theory, optimization, and application literature [13]. Nest asks whether a returned tree is locally optimal under an exact topology rotation and uses the answer as both a diagnostic and an optimizer.

Graph hierarchical clustering. Paris supplies an inexpensive agglomerative graph hierarchy related to modularity [2]. Louvain optimizes modularity [1]; we score its final partition as a two-level tree. Dasgupta's cost and its revenue dual have inspired hierarchical algorithms with guarantees [4, 11]. Because these objectives differ from structural entropy, optimizing Eq. (1) requires its own local identity.

Exact inference and heuristic accuracy. Cluster trellises support exact hierarchical inference [9]. Our routine is an SE specialization: the prior binary-sufficiency property justifies the subset recurrence, while our arbitrary-subset formulation differs from SuperTAD's contiguity-constrained dynamic programs.

Local search in tree space. NNI is a classical neighborhood for evolutionary, or phylogenetic, trees; its reconfiguration distance is itself nontrivial [8]. Recent work studies NNI local search for hierarchical-clustering costs [5]. PERCH uses purity-enhancing rotations during online tree construction, while GRINCH combines rotations with longer-range grafts for general linkage functions [6, 10]. Our formula instead targets weighted graph structural entropy and exposes the two cross-module weights controlling an NNI rotation. Unlike unit-clique landscape analyses, our method evaluates the complete objective on arbitrary weighted graphs. Its new content is the verified optimizer, compound search, multi-start construction, comparative evaluation, and exact global audit.

6 Conclusion

We introduced Nest, which derives an exact weighted NNI identity and uses it for monotone refinement, a local-optimality certificate, and safe bounded escape from shallow one-step traps. On the frozen five-regime suite, Nest achieved the lowest mean entropy in every regime and outperformed both HCSE and BBM on all 500 paired graphs. Prior binary sufficiency together with our exact arbitrary-subset dynamic program shows that 32-start Nest reaches the global optimum on 370/375 sealed instances across $n = 12, 14, 16$, versus 2/375 for HCSE and 1/375 for label-free BBM. The operator audit further shows that hierarchy construction, local certification, and exact optimality provide complementary views of search quality. Together, the local identity, certified search, and exact audit form a reproducible framework for optimizing structural-entropy hierarchies.

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A Proof Details and Additional Basin Analysis

This appendix is included in the single review PDF. It supplies proofs omitted from the 12-page main paper and additional finite-instance evidence; none of its claims is needed to interpret the main experimental tables.

Proof of Theorem 1. First suppose the three child modules have positive volume; the zero-volume case follows by deleting modules with no incident positive-weight edge and taking the zero-contribution limit. Only the terms for A, B, C, X, and Y can change. Omitting the common factor $1/\operatorname{vol}(V)$, their old and new contributions are

$$L_{\text{old}} = g_A \log_2 \frac{V_X}{V_A} + g_B \log_2 \frac{V_X}{V_B} + g_X \log_2 \frac{V_P}{V_X} + g_C \log_2 \frac{V_P}{V_C},$$
$$L_{\text{new}} = g_A \log_2 \frac{V_P}{V_A} + g_B \log_2 \frac{V_Y}{V_B} + g_C \log_2 \frac{V_Y}{V_C} + g_Y \log_2 \frac{V_P}{V_Y}.$$

Using $g_X = g_A + g_B - 2W(A,B)$ and $g_Y = g_B + g_C - 2W(B,C)$, the coefficients of g_A, g_B, g_C telescope to zero. The remaining terms are

$$2W(A,B)\log_2(V_P/V_X) + 2W(B,C)\log_2(V_Y/V_P).$$

Restoring $1/\operatorname{vol}(V)$ proves Eq. (2). External boundary weights and $W(A, C)$ do not enter the delta.

Proof of Proposition 1. Delete X's codeword and substitute $g_X = g_A + g_B - 2W(A, B)$ in Eq. (1); this gives Eq. (3). The symmetric calculation for inserting Y gives Eq. (4).

Proof of Proposition 2. Only moves with $\Delta H^T < -\varepsilon$ are committed, and their endpoint is independently verified. At termination every enumerated legal move has $\Delta H^T \geq -\varepsilon$.

Proof of Proposition 3. A pair is committed only after its final tree is fully recomputed and found to have lower entropy than the tree preceding the pair. The first move is not committed independently, so its sign cannot weaken the output guarantee. A two-edge path may have a positive first delta and a negative net delta; the regression witness in Sec. 4 instantiates this case.

Proof of Proposition 4 (included for completeness). At a node P, choose two children with disjoint leaf sets A, B, insert their union $S = A \cup B$ below P, and make A, B children of S. The objective change is

$$-\frac{2W(A, B)}{\operatorname{vol}(V)} \log_2 \frac{V_P}{V_S} \leq 0.$$

Repeated insertion converts every multiway node to binary without increasing the objective.

A.1 Exact and sampled basin probabilities

The finite hit rates in the main paper do not supply a per-start probability. For a fixed graph G, deterministic refinement map F, optimum H^*, and start measure μ, that probability is

$$p_G = \sum_T \mu(T)\, \mathbf{1}\{H^{F(T)}(G) = H^*(G)\}.$$

It depends on the graph, start distribution, tie-breaking, and refinement configuration. The exact subset DP supplies H^* but not p_G.

For $n \leq 8$ we enumerate all $(2n-3)!!$ unordered rooted binary labeled topologies. Uniform topology mass is not NEST's restart distribution. Under the pairwise coalescent, a topology with child subtrees of a, b leaves has compatible-history count

$$h(T) = \binom{a+b-2}{a-1} h(T_A) h(T_B), \qquad \mu(T) = \frac{h(T)}{\prod_{k=2}^{n} \binom{k}{2}}.$$

The history counts sum exactly to the denominator. Table 6 shows that the selected hard $n = 8$ graph has exact coalescent basin mass $p_G = 0.252365$; because

Table 7. Sealed, budget-matched exact audit on 250 independently generated 12-vertex HSBMs. Each restarted constructor receives 32 candidate calls; the lowest structural entropy is selected without consulting H^*. Valid reports successful instances. Exact rates use all 250 graphs as denominator. Gap is mean relative entropy gap (%) ± 95% CI over valid instances.

Method	Calls	Valid	Exact	Gap (%)	Time (s)
NEST (coalescent)	32	250/250	**249 (99.6%)**	0.00031958±0.000629	0.171
HCSE	32	250/250	2 (0.8%)	10.902±0.971	0.024
BBM (oracle k)	32	248/250	1 (0.4%)	15.054±1.13	0.402
BBM (label-free)	32	250/250	1 (0.4%)	10.237±0.593	0.299
SE agglomerative	1	250/250	44 (17.6%)	1.7636±0.268	0.00027

the standard pool misses it, 32 independent starts give $1-(1-p_G)^{32}=0.999909$, not certainty.

Table 6. Exact basin mass on one fixed noisy HSBM graph at each size. p_U uses the uniform measure over unordered rooted binary labeled topologies; p_C uses NEST's pairwise-coalescent measure. "Strict" requires every planted fine and coarse block to occur as a clade.

n	starts	pool hits?	$p_U(H^*)$	$p_C(H^*)$	p_C(strict)
5	105	yes	100.000%	100.000%	0.000%
6	945	yes	100.000%	100.000%	0.000%
7	10,395	yes	100.000%	100.000%	0.000%
8	135,135	no	24.848%	25.237%	0.000%

A.2 Detailed 12-vertex method audit

Table 7 retains the full method and runtime breakdown for the largest exact-audit stratum. The main paper reports the cross-size comparison.

At $n=12$, exact enumeration would require 13,749,310,575 topologies. Table 8 therefore uses direct Bernoulli sampling rather than the earlier first-hit diagnostic. The estimates vary substantially by graph: 8.07%–46.43% per start. On noisy-11 this implies only 93.23% for 32 random starts (a conservative 95% lower value of 91.87%), explaining why the observed finite-suite success must not be read as a probabilistic guarantee.

Table 8. Direct Monte Carlo basin estimates on the eight preidentified hard $n = 12$ graphs (10,000 independent coalescent starts each). Intervals are exact 95% Clopper–Pearson intervals. $q_{32} = 1 - (1 - \hat{p})^{32}$; its lower column uses the lower confidence limit for p. Strict recovery was observed 0/10,000 times on every graph (per-graph 95% upper bound 0.0369%).

Graph	$\hat{p}(H^*)$	95% CI	q_{32}	lower q_{32}
noisy-0	17.19%	[16.46, 17.94]%	99.761%	99.683%
noisy-8	14.61%	[13.92, 15.32]%	99.362%	99.175%
noisy-9	20.56%	[19.77, 21.37]%	99.937%	99.913%
imbal.-0	46.43%	[45.45, 47.41]%	100.000%	100.000%
weak-9	30.61%	[29.71, 31.52]%	99.999%	99.999%
noisy-11	8.07%	[7.54, 8.62]%	93.229%	91.872%
noisy-17	17.75%	[17.01, 18.51]%	99.808%	99.743%
weak-17	17.86%	[17.11, 18.63]%	99.816%	99.754%

Objective success is deliberately separated from strict planted recovery. No one of the 80,000 hard-case endpoints contained every planted fine and coarse block as a clade; the exact $n = 8$ enumeration likewise gives zero mass. This strict event is stronger than dendrogram purity and exposes objective– recovery misalignment on these small realized HSBMs. Thus the restart evidence supports optimization of H^T, not a claim that every entropy optimum equals the generator's planted hierarchy.